Divergent terms

1 Group I (with c.c.)

1.1 Row I

NA

1.2 Row II

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times$$
$$\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right] \tag{1.1}$$

1.3 Row III

NA

1.4 Row IV

NA

1.5 Row V

First part

NA

Second part

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\left[2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{1.2}$$

2 Adding Group I

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times$$
$$\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\left[4\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{3}\right]\int_{x,w,x',w'} e^{-ik\cdot(w'-w)}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{ab}(x)-U^{ab}(w)\Big]\rho^{b'}(x')\,\rho^b(x). \tag{2.1}$$

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times$$
$$\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\left[4\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{3}\right]\int_{x,w,x',w'} e^{-ik\cdot(w'-w)}\left[-\frac{2}{\epsilon}-2\gamma_E-\log\frac{(x-w)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}$$
$$\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{ab}(w)-U^{ab}(x)\Big]\rho^{b'}(x')\,\rho^b(x). \tag{2.2}$$

Simplifying, we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu_{\overline{MS}}^2}{4}\right)\right]\times$$
$$\left[4\,\log\left(1-\frac{\Lambda}{k^+}\right)+\log\left(1-\frac{k^+}{\vee}\right)\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right] \tag{2.3}$$

Setting $\Lambda\to 0$ and $\vee\to\infty$, we get

$$\frac{d^3N_{\rm NLO}}{d^3k}=0$$

3 Group II

3.1 Row I

3.1.1 First part

NA

3.1.2 Second part

NA

3.2 Row II

NA

3.3 Row III

NA

3.4 Row IV

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]$$
$$\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\left[\log\frac{(\vee-k^+)\,(k^+)}{\vee\,\Lambda}\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.$$
$$\left.-\,\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)$$
$$+\,\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]$$
$$\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x) \tag{3.1}$$

Here the plus prescription in the first term is the *locally subtracted* one on the integration region, $\int_{k^+/\vee}^1 d\xi\,\frac{h(\xi)}{(1-\xi)_+}\equiv\int_{k^+/\vee}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}$; the endpoint it removes is written out explicitly as the second (blue) term, with $\log\frac{(\vee-k^+)k^+}{\vee\Lambda}=\log\frac{k^+}{\Lambda}+\log\left(1-\frac{k^+}{\vee}\right)$. The extraction of the JIMWLK, BFKL and finite parts is carried out in Section 5.

4 Group III (with c.c.)

4.1 Row I

NA

4.2 Row II

NA

4.3 Row III

4.3.1 First part

NA

4.3.2 Second part

NA

4.4 Row IV

NA

4.5 Row V

NA

4.6 Row VI

NA

5 Group II, Row IV

We start from Eq. (3.1). Throughout we use $d_\perp = 2-\epsilon$, so that

$$\left[-\frac{d_\perp}{\epsilon} - d_\perp\,\gamma_E - \frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1\right] = -\frac{2}{\epsilon} + 1 - 2\,\gamma_E - \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) - 1$$
$$= -\frac{2}{\epsilon} - 2\,\gamma_E - \log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right) \equiv -L_\epsilon\,, \tag{5.1}$$

and the leading order cross section is

$$\frac{d^3N_{\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\times\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\,, \tag{5.2}$$

with its ξ-rescaled version $\frac{d^3N_{\xi\,\rm LO}}{d^3k}$ carrying the phase $e^{-ik\cdot(y'-y)/\xi}$.

5.1 Extraction of the JIMWLK term

Splitting the blue logarithm of the second term of Eq. (3.1), $\log\frac{(\vee-k^+)\,k^+}{\vee\,\Lambda} = \log\frac{k^+}{\Lambda} + \log\left(1-\frac{k^+}{\vee}\right)$, its $\log\frac{k^+}{\Lambda}$ part is the JIMWLK contribution of this row,

$$\frac{d^3N^{\rm JIMWLK}_{\text{G-II, R IV}}}{d^3k} = \frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,[-L_\epsilon]$$
$$\left[U^{ce'}(y') - U^{ce'}(x')\right]\times\left[U^{ce}(y) - U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\,, \tag{5.3}$$

which is moved to the two-ρ evolution (JIMWLK) file, where it appears verbatim with the bracket $\left[-\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4} - 1\right] = -L_\epsilon$. The $\log\left(1-\frac{k^+}{\vee}\right)$ part of the blue logarithm stays here. In the third term of Eq. (3.1) we may now set $\Lambda\to 0$ in the upper limit, $\frac{k^+}{k^++\Lambda}\to 1$ (the integrand is regular at $\xi=1$), and $d_\perp\to 2$ (no pole multiplies it).

5.2 Assembling the splitting function

With the *standard* plus prescription on $[0,1]$,

$$\int_a^1 d\xi\,\frac{h(\xi)}{(1-\xi)_+} = \int_a^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi} + h(1)\,\log(1-a)\,, \qquad a = \frac{k^+}{\vee}\,, \tag{5.4}$$

the first term of Eq. (3.1) and the leftover $\log\left(1-\frac{k^+}{\vee}\right)\left[-L_\epsilon\right]$ endpoint term combine into a single ξ-integral with $\frac{\xi}{(1-\xi)_+}$ in the standard convention. Denoting

$$\mathcal{P}_{gg,+}(\xi) = 2N_c\left[\xi\,(1-\xi) + \frac{(1-\xi)}{\xi} + \frac{\xi}{(1-\xi)_+}\right], \qquad \mathcal{P}_{gg}(\xi) = \mathcal{P}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi)\,, \qquad \beta_0 = \frac{11\,N_c}{3}\ \ \text{(pure glue)}\,, \tag{5.5}$$

and adding and subtracting the $\frac{\beta_0}{2}\delta(1-\xi)$ piece, we obtain

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2}{8\pi^2}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}L_\epsilon\times$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{g^2\,\beta_0}{16\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}L_\epsilon\times$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)$$
$$+\frac{g^2}{4\pi^2}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]\times$$
$$\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\,\rho^{e'}(x')\rho^e(x)\,. \tag{5.6}$$

The violet and green terms are the only carriers of L_ϵ; the $\frac{\beta_0}{2}\delta(1-\xi)$ piece inside $\mathcal{P}_{gg}$ cancels the green (running-coupling) term identically, so keeping both is an exact add-and-subtract that makes the coupling renormalization explicit.

5.3 The soft subtraction and the extraction of the BFKL term

The $2N_c\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$ in the violet term produces a hidden $\log\frac{\vee}{k^+}$ from $\xi\to 0$: there the parent positions collapse onto the sibling, $y=\xi w+(1-\xi)z\to z$, $y'=\xi w'+(1-\xi)z\to z$, so a rapidity logarithm with an L_ϵ coefficient would appear, which is not physical (the coefficient of $\log\vee$ must be the μ-independent BFKL kernel). Its $\xi\to 0$ limit is subtracted *exactly* with

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{1}{\pi}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}$$
$$\times\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\,, \tag{5.7}$$

which is IR and UV finite in $d=2$, contains no μ, and is symmetric under $(x,y)\leftrightarrow(x',y')$. Writing $W(u,u')=[U^{ce'}(u')-U^{ce'}(x')][U^{ce}(u)-U^{ce}(x)]\rho^{e'}(x')\rho^e(x)$ and $K(a,b,z)=\frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}$, and using $\int_z^{(d=2-\epsilon)}K(y,y',z)=-\pi L_\epsilon$, the combination

$$2N_c\frac{1-\xi}{\xi}\Big[L_\epsilon\frac{d^3N_{\xi\,\rm LO}}{d^3k}+\frac{d^3N_\xi^{\rm sub}}{d^3k}\Big] = -\frac{2N_c(1-\xi)}{\xi}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K$$
(5.8)

contains no $\frac{2}{\epsilon}$ and no μ, and at fixed w,w' the square bracket is $O(\xi)$ as $\xi\to 0$ (since $y-z=\xi(w-z)$), so no $\log\vee$ survives with an L_ϵ coefficient. The exact value of the subtraction follows from the change of variables $y=z+\xi(w-z)$, $y'=z+\xi(w'-z)$ (Jacobian ξ^4, $K(y,y',z)=K(w,w',z)/\xi^2$, phase $\to e^{-ik(w'-w)}$) and $\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}=\log\frac{\vee}{k^+}-\big(1-\frac{k^+}{\vee}\big)$:

$$-\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\rm sub}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{k^+}-\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}K(w,w',z)\,K(x,x',z)$$
$$\times\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\,. \tag{5.9}$$

We therefore add and subtract Eq. (5.9): the unintegrated subtraction joins the violet term (rendering its soft sector μ-free), while of its exact value the $\log\frac{\vee}{k^+}$ part is the BFKL contribution of this row,

$$\frac{d^3N^{\rm BFKL}_{\text{G-II, R IV}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-$$
(5.10)

moved to the BFKL file (it appears there verbatim, G-II Row IV), and the $-\big(1-\frac{k^+}{\vee}\big)$ part remains below as the boundary term. Note the coefficient of $\log\frac{\vee}{k^+}$ is manifestly UV finite and μ-independent, as a BFKL kernel must be.

5.4 The finite terms

Collecting everything, the part of Row IV that belongs to the finite ("other") terms is

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2}{8\pi^2}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\times\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,\big[U^{ce}(z)-U^{ce}(x)\big]\,\rho^{e'}(x')\,\rho^{e}(x)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{g^2\,\beta_0}{16\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\times\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\times\big[U^{ce}(z)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]\times\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x)\,,
\end{aligned}
\tag{5.11}
$$

with $\mathcal{P}_{gg}$ in the *standard* $[0,1]$ plus-prescription convention. Summary of the bookkeeping:

(i) $\big(\text{second term}\big)+\big(\text{fourth term}\big)+\big(\text{BFKL entry}\big)=0$ identically — the construction is an exact add-and-subtract; all three objects are UV finite, and the physical origin of the BFKL logarithm is the $\xi\to 0$ endpoint of the violet term.

(ii) The $\frac{\beta_0}{2}\delta(1-\xi)$ piece of the violet term cancels the green term identically; the soft $2N_c\frac{1-\xi}{\xi}$ piece of the violet term combines with the second term into the μ-free difference form above. The only surviving UV logarithms are $L_\epsilon\otimes 2N_c\big[\frac{\xi}{(1-\xi)_+}+\xi(1-\xi)\big]$ — the physical collinear (DGLAP) kernel of the measured gluon, including its $(1-\xi)_+$ endpoint $\log\big(1-\frac{k^+}{\vee}\big)$ piece — and the green running-coupling term. These are to be combined with the Group-I ultraviolet terms of the "Adding Group I" section in the final pole accounting ($\beta_0=\frac{11N_c}{3}$: coupling renormalization plus the collinear subtraction for the produced gluon).

(iii) Together with the JIMWLK and BFKL entries extracted above, Eq. (5.11) reproduces the starting expression (3.1) identically, to $O(\Lambda)$ and $O(\epsilon)$.